EXPERIMENT – 1

1. A class consists of 40 students who have secured different grades in an examination. The grades are classified as F, E, D, C, B, A, and S, where F represents the lowest grade and S represents the highest grade.

The number of students who secured each grade is given below.

Grade	Number of Students
F	3
E	4
D	5
C	8
B	9
A	7
S	4
Total	40

Develop a Python program to perform the following tasks:

1. Store the grades and corresponding number of students in suitable data structures.

2. Draw a pie chart to represent the percentage distribution of students across the seven grades.

3. Display the grade labels and percentage values on the pie chart.

4. Draw a bar chart to compare the number of students who secured each grade.

5. Label the X-axis as Grades and the Y-axis as Number of Students.

6. Display the exact number of students above each bar.

7. Add suitable chart titles, legends, and axis labels.

8. Identify the grade secured by the maximum and minimum number of students.

9. Display the percentage of students who secured grades A and S combined.

Aim

To write a Python program to represent the distribution of students across grades using a pie chart and a bar chart, identify the maximum and minimum grade counts, and calculate the combined percentage of students who obtained grades A and S.

Procedure

1. Import the Matplotlib library for creating charts.
2. Store the grade labels and their corresponding student counts in lists.
3. Calculate the total number of students and the percentage for each grade.
4. Draw a pie chart with grade labels, percentage values, a title, and a legend.
5. Draw a bar chart with Grades on the X-axis and Number of Students on the Y-axis.
6. Display the exact count above each bar and add a chart title and legend.
7. Find the maximum and minimum student counts and identify their corresponding grades.
8. Calculate and print the combined percentage for grades A and S.

Python Code

```
import matplotlib.pyplot as plt

grades = ['F', 'E', 'D', 'C', 'B', 'A', 'S']
students = [3, 4, 5, 8, 9, 7, 4]
total = sum(students)

percentages = [n / total * 100 for n in students]

plt.figure(figsize=(7, 5))
plt.pie(students, labels=grades, autopct='%1.1f%%', startangle=90)
plt.title('Percentage Distribution of Grades')
plt.legend(grades, title='Grades')
plt.show()

plt.figure(figsize=(7, 5))
bars = plt.bar(grades, students, label='Students')
plt.xlabel('Grades')
plt.ylabel('Number of Students')
plt.title('Number of Students in Each Grade')
plt.bar_label(bars)
plt.legend()
plt.show()

max_count = max(students)
min_count = min(students)
max_grades = [g for g, n in zip(grades, students) if n == max_count]
min_grades = [g for g, n in zip(grades, students) if n == min_count]

as_percentage = (students[5] + students[6]) / total * 100

print("Total Students:", total)
print("Maximum Students Grade:", ', '.join(max_grades), "-", max_count)
print("Minimum Students Grade:", ', '.join(min_grades), "-", min_count)
print("Percentage of A and S Grades Combined:", round(as_percentage, 2), "%")
```

Output Screenshots

Figure 1: Pie chart showing the percentage distribution of grades.

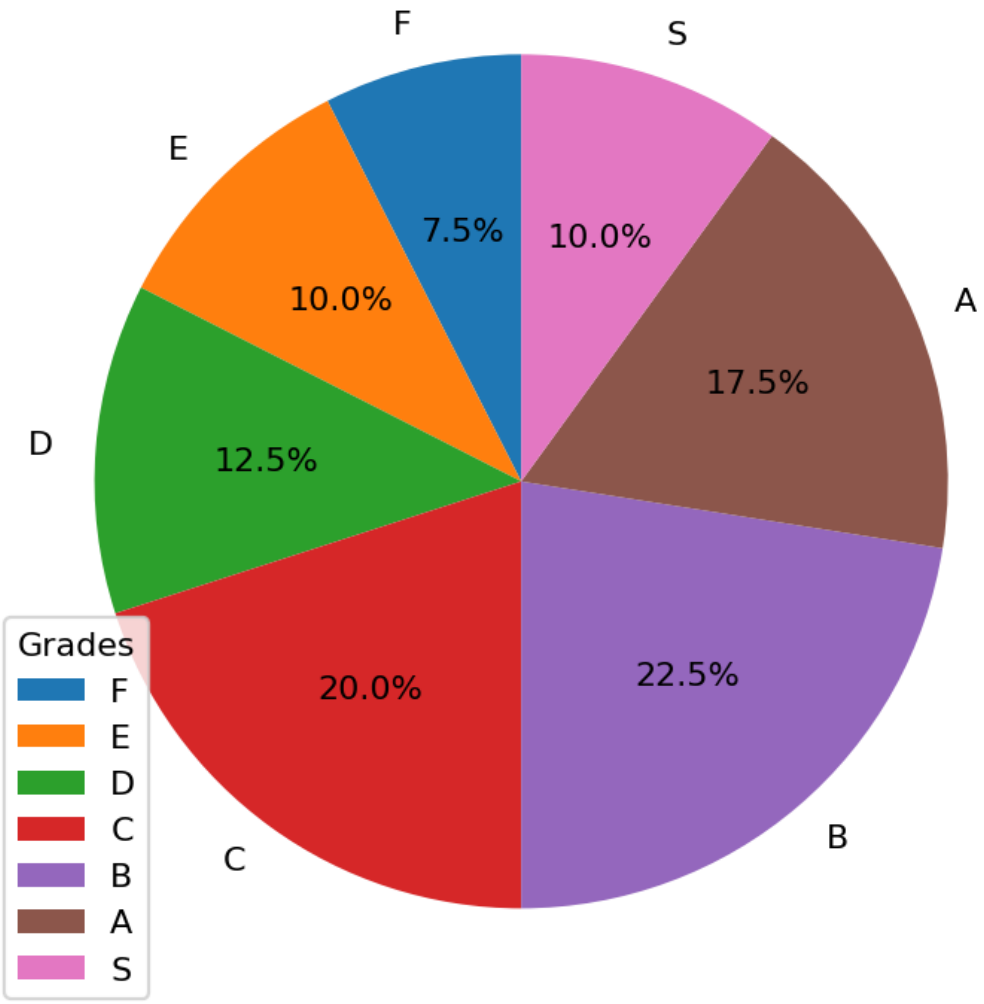
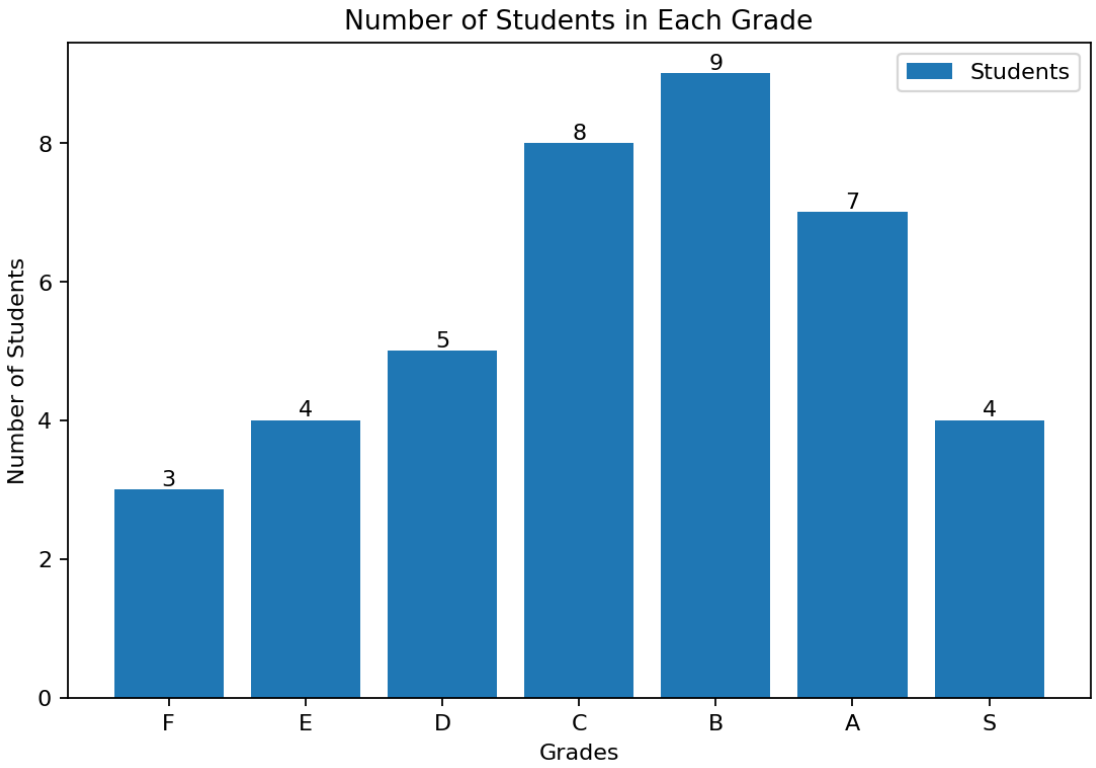

Percentage Distribution of Grades
F
S
E
A
D
B
C
7.5%
10.0%
10.0%
17.5%
12.5%
22.5%
20.0%
Grades
F
E
D
C
B
A
S

Figure 2: Bar chart comparing the number of students in each grade.



Console Output

Total Students: 40
Maximum Students Grade: B - 9
Minimum Students Grade: F - 3
Percentage of A and S Grades Combined: 27.50 %

Result

The program successfully visualizes the grade distribution using a pie chart and a bar chart. Grade B has the highest count with 9 students. Grade F has the lowest count with 3 students. Grades A and S together account for 27.50% of the class.